Physical Layer: Guided Transmission media: Magnetic Media, Twisted pairs, Coaxial Cable, Fiber Optics, Unguided Transmission Media: Radio waves, Microwaves, Infrared.

Guided Transmission media

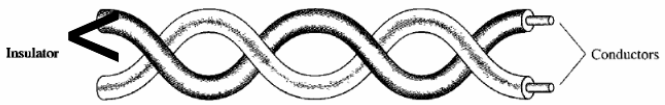
The purpose of the physical layer is to transport bits from one machine to another Media are roughly grouped into guided media, such as copper wire and fiber optics.

Magnetic Media

One way to transfer large amounts of data is by copying it onto **magnetic tapes or removable media (such as DVDs)** and physically transporting them to the destination. Although this method is slower than electronic communication, it is often **more cost-effective and provides extremely high bandwidth** for transferring massive datasets. For example, a box containing **1000 Ultrium tapes** can store **800 terabytes (6.4 petabits)** of data. If delivered within **24 hours**, it provides an effective bandwidth of about **70 Gbps**, and if transported within **1 hour**, the bandwidth exceeds **1700 Gbps**, which is higher than most computer networks. Therefore, for very large data transfers, physical transportation of storage media can be an efficient and economical solution.

Twisted pairs

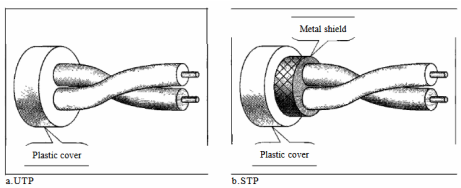
Twisted pair cable transmits data in the form of electrical signals A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown in Fig.



One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference. A signal is usually carried as the difference in voltage between the two wires in the pair. This provides better immunity to external noise because the noise tends to affect both wires the same, leaving the differential unchanged.Twisting of wires minimizes electromagnetic interference (EMI) and crosstalk. Suitable for Ethernet networks with speeds up to 10 Gbps

Types of Twisted Pair Cables

1. **UTP (Unshielded Twisted Pair)**
    - No additional shielding.
    - Less expensive and easy to install.
    - Commonly used in Ethernet LANs.
2. **STP (Shielded Twisted Pair)**
    - Includes shielding around the wires.
    - Reduces external interference and crosstalk.
    - Used in environments with high electromagnetic noise.



a.UTP b.STP

Advantages of Twisted Pair Cable

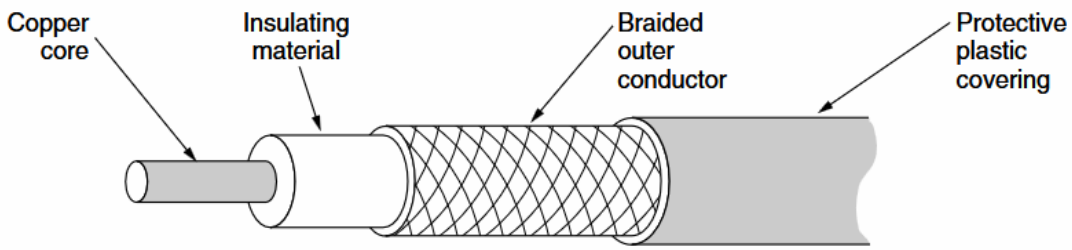
1. **Low Cost** – Less expensive than coaxial and fiber optic cables.
2. **Easy Installation** – Lightweight, flexible, and simple to install.
3. **Widely Used** – Commonly used in telephone systems and Local Area Networks (LANs).

Disadvantages of Twisted Pair Cable

1. **Limited Distance** – Effective transmission is limited to about **100 meters**.
2. **Lower Bandwidth** – Offers lower bandwidth than fiber optic cables.
3. **Susceptible to Noise** – Unshielded Twisted Pair (UTP) can be affected by electromagnetic interference.
4. **Signal Attenuation** – Signal strength decreases over longer distances.

Coaxial Cable

A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh. The outer conductor is covered in a protective plastic sheath .



Structure

1. **Copper Core** – Carries the electrical signal.
2. **Insulating Material** – Separates the core from the shield.
3. **Braided Outer Conductor** – Prevents interference and signal leakage.
4. **Protecting plastic covering** – Protects the cable from physical damage

Features

- Higher bandwidth than twisted pair cable.
- Better protection against electromagnetic interference.
- Supports both analog and digital signals.
- Can transmit data over longer distances.
- More durable and reliable.

Two kinds of coaxial cable are widely used. One kind, 50-ohm cable, is commonly used when it is intended for digital transmission from the start. The other kind, 75-ohm cable, is commonly used for analog transmission and cable television.

Advantages of Coaxial Cable

1. **Higher Bandwidth compared to twisted pair cable**
2. **Better Noise Immunity** – The shielding protects the signal from electromagnetic interference (EMI).
3. **Longer Transmission Distance** – Can transmit data over longer distances without significant signal loss.
4. **Suitable for Multiple Applications** – Used in cable TV, broadband Internet, CCTV, and radio communication.

Disadvantages of Coaxial Cable

1. **More Expensive than twisted pair cable**– Costs more than twisted pair cables.
2. **Bulky and Less Flexible** – Thicker cable makes installation and routing more difficult.
3. **Difficult Installation** – Requires special connectors and careful installation.
4. **Lower Speed than Fiber Optic** – Provides lower bandwidth and speed compared to fiber optic cables.

Fiber Optics

A fiber optic cable is a guided transmission medium made of glass or plastic that transmits data in the form of light signals instead of electrical signals. It provides very high bandwidth, high-speed communication, and is immune to electromagnetic interference.

The working of an optical fiber is based on the behavior of light. Light travels in a straight line through a uniform medium, but when it passes from one medium to another with a different density, it changes direction due to refraction. In optical fibers, light is guided through the fiber core by the principle of Total Internal Reflection (TIR), allowing it to travel long distances with very little signal loss.

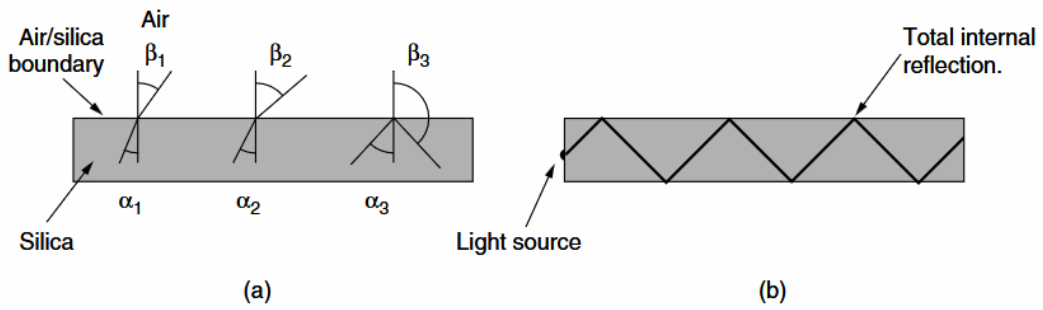


Figure 2-6. (a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles. (b) Light trapped by total internal reflection.

Transmission of Light Through Fiber : Optical fibers are made of glass, which, in turn, is made from sand, an inexpensive raw material available in unlimited amounts. Glassmaking was known to the ancient Egyptians, but their glass had to be no more than 1 mm thick or the

Fiber Cables: Fiber optic cables are similar to coax, except without the braid. Figure shows a single fiber viewed from the side. At the center is the glass core through which the light propagates. In multimode fibers, the core is typically 50 microns in diameter, about the thickness of a human hair. In single-mode fibers, the core is 8 to 10 micron

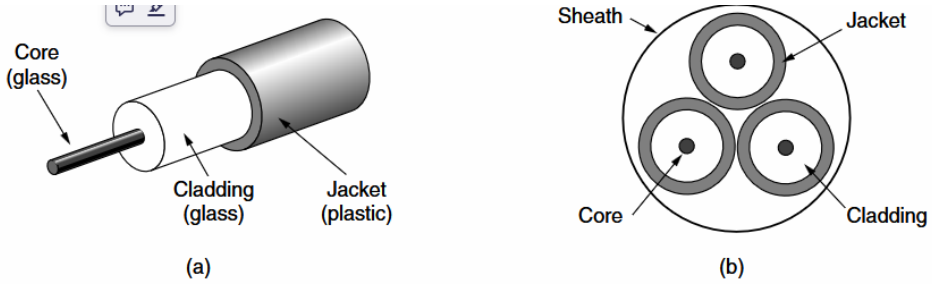
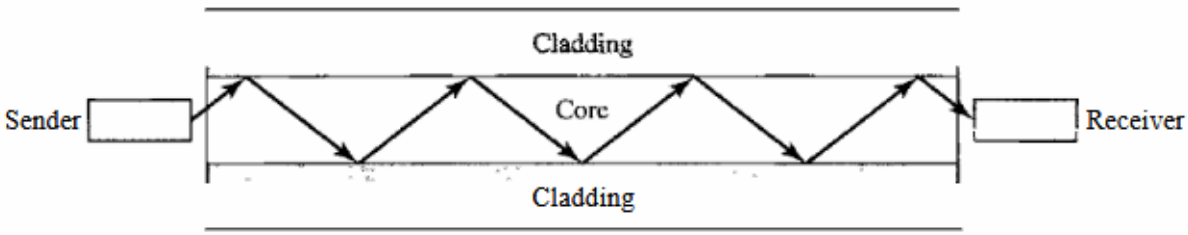


Figure 2-8. (a) Side view of a single fiber. (b) End view of a sheath with three fibers.

The core is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core. Next comes a thin plastic jacket to protect the cladding. Fibers are typically grouped in bundles, protected by an outer sheath. Two kinds of light sources are typically used to do the signaling. These are LEDs (Light Emitting Diodes) and semiconductor lasers.

The receiving end of an optical fiber consists of a photodiode, which gives off an electrical pulse when struck by light. The response time of photodiodes, which convert the signal from the optical to the electrical domain, limits data rates to about 100 Gbps.



Advantages and Disadvantages of Optical Fiber:

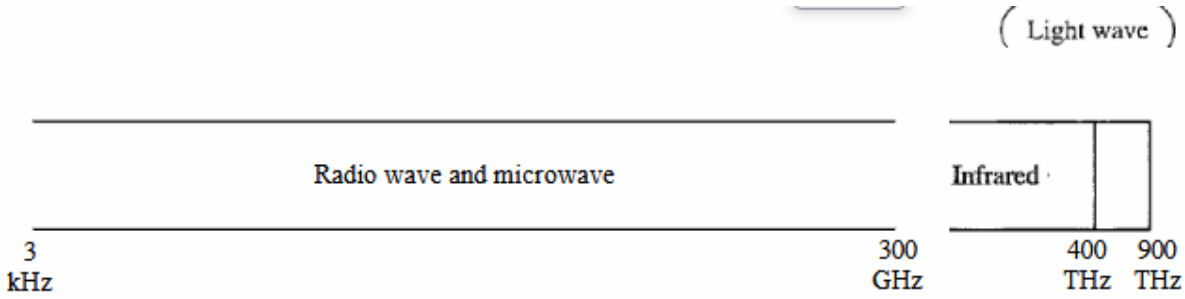
Advantage	Description
High Bandwidth	Supports very high data transmission rates and bandwidth.
Low Signal Attenuation	Signals can travel up to **50 km** without regeneration, reducing the need for repeaters.
Immunity to Electromagnetic Interference (EMI)	Light signals are not affected by electrical or electromagnetic noise.
Corrosion Resistance	Glass fibers are resistant to corrosion and environmental damage.
Light Weight	Fiber-optic cables are thinner and lighter than copper cables, making installation easier.
High Security	Difficult to tap, providing better data security than copper cables.

Disadvantage	Description
Difficult Installation and Maintenance	Requires skilled technicians and specialized equipment.
Unidirectional Communication	A single fiber carries light in one direction only; two fibers are required for two-way communication.
High Cost	Fiber cables and associated devices are more expensive than copper cables.

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

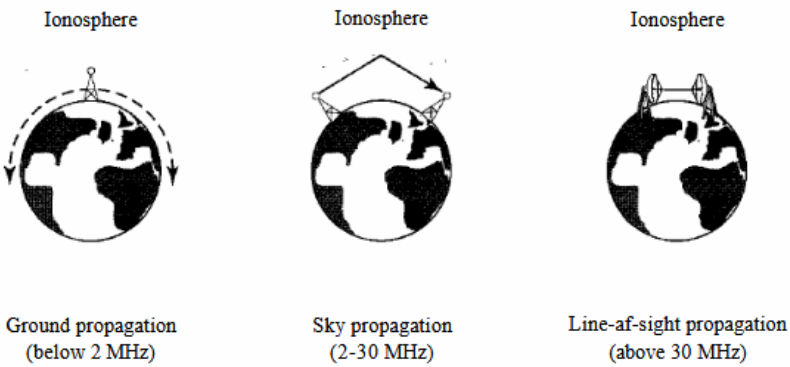
Electromagnetic Spectrum for Wireless Communication

The **electromagnetic (EM) spectrum** is the range of electromagnetic waves used to transmit information without physical cables. Different frequency bands are used for different wireless communication applications.



Band	Frequency Range	Applications
Radio Waves	3 kHz – 300 MHz	AM/FM radio, TV broadcasting, maritime and aviation communication
Microwaves	300 MHz – 300 GHz	Mobile networks, Wi-Fi, Bluetooth, satellite communication, radar
Infrared (IR)	300 GHz – 400 THz	TV remote controls, short-range wireless communication
Visible Light	400 THz – 790 THz	Fiber-optic communication, Li-Fi (Light Fidelity)

Unguided signals can travel from the source to destination in several ways: ground propagation, sky propagation, and line-of-sight propagation, as shown in Fig



1. **Ground Propagation** : Radio waves travel along the surface of the Earth.Used for low-frequency (LF) and medium-frequency (MF) signals. Suitable for AM radio broadcasting and maritime communication.
2. **Sky Propagation** : Radio waves are reflected by the ionosphere back to Earth. Used for high-frequency (HF) communication. Enables long-distance communication beyond the horizon. Applications: International radio broadcasting, amateur radio.
3. **Line-of-Sight (LOS) Propagation**: Signals travel in a straight line from transmitter to receiver. Requires an unobstructed path. Used for very high frequencies (VHF, UHF, microwaves). Applications: TV broadcasting, mobile communication, Wi-Fi, satellite communication.

Microwaves : Microwaves are electromagnetic waves with frequencies ranging from **1 GHz to 300 GHz**. They are widely used for **wireless communication** because they support **high-speed data transmission** over long distances.

Characteristics

- **Frequency Range:** 1 GHz – 300 GHz.
- **Unidirectional:** Signals travel in a narrow beam, so transmitting and receiving antennas must be properly aligned.
- **Line-of-Sight Propagation:** A clear path is required between antennas; repeaters are needed for long distances.
- **Cannot Penetrate Walls:** Very high-frequency microwaves cannot pass through buildings or large obstacles.
- **High Bandwidth:** Supports high data rates because of its wide frequency range.
- **Licensed Spectrum:** Some microwave frequency bands require government authorization for use.

Applications

- Satellite communication
- Mobile communication (4G/5G)
- Wi-Fi (2.4 GHz and 5 GHz)
- Radar systems
- Point-to-point communication links
- Microwave ovens (industrial and domestic)

Advantages

- Supports **high-speed data transmission**.
- High bandwidth and large communication capacity.
- Suitable for long-distance wireless communication.
- Narrow beam reduces interference between different communication links.

Disadvantages

- Requires **line-of-sight** communication.
- Cannot easily penetrate walls and obstacles.
- Repeaters are needed for long-distance communication.
- Some frequency bands require government licensing.

Infrared (IR)

Infrared (IR) waves are electromagnetic waves with frequencies ranging from **300 GHz to 400 THz** (wavelengths **1 mm to 770 nm**). They are mainly used for **short-range wireless communication** between electronic devices.

Characteristics

- **Frequency Range: 300 GHz – 400 THz**.
- Used for **short-range communication**.
- **Cannot penetrate walls**, preventing interference between adjacent rooms.
- Uses **line-of-sight propagation**, so devices must face each other.
- **Sunlight contains infrared rays**, which can interfere with outdoor communication.
- Provides **high bandwidth** and supports high data rates.

Applications

- TV and air-conditioner remote controls
- Wireless keyboards and mice
- Communication between PCs and printers (IrDA)
- Short-range data transfer between electronic devices

Advantages

- High data transmission speed.
- Secure communication due to limited range.
- License-free communication.

Disadvantages

- Very short communication range (typically up to **8 m**).
- Requires **line-of-sight** between devices.
- Cannot penetrate walls or obstacles.
- Not suitable for outdoor communication because sunlight can cause interference.